

Samitha Ranasinghe

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EXECUTIVE SUMMARY

I am looking for an coop opportunity in Robotics and Embedded Engineering. I am able to work starting May 11th and would be able to work until December with a small break beginning of August.

EDUCATION

M.S. Computer Engineering Purdue University – West Lafayette, IN <i>Focus in Embedded Systems and Robotics</i>	Jan. '25 – May '27 GPA: 3.46
B.S. Computer Engineering Purdue University – West Lafayette, IN <i>Concentration in AI/ML and Automatic Control</i>	Aug. '20 – Dec. '24 GPA: 3.71

PROJECTS

Embedded RF Receiver Reverse Engineering - Utilizing a ESP32 to intercept and decode 2.4GHz RF signals from a proprietary Logitech wireless keyboard missing its original dongle. - Reverse-engineering the HID report descriptors and radio protocol layers to establish a secure, low-latency connection via Bluetooth.	Mar. 2026 – Present
Autonomous Mapping and Search & Rescue Simulation - Integrated SLAM (Simultaneous Localization and Mapping) with an A* pathfinding global planner to enable autonomous exploration of unknown environments in Gazebo. - Developed a multi-threaded ROS node utilizing OpenCV (cv2) for real-time color segmentation and object detection to identify and track target "victims" (colored balls) while navigating obstacles.	Sep. 2025 – Dec. 2025
USB Transmitter & Data Buffer <i>ASIC Design</i> - Designed a USB Full-Speed Bulk Transfer Endpoint integrated as an AHB SoC module. - Developed RTL for packet framing, synchronization, and endpoint handshaking to ensure protocol compliance. - Verified hardware functionality through rigorous testbenches focused on timing closure and bus integration.	Mar. 2025 – May 2025
ESP32 Wearable Evaluation Board <i>PCB Design</i> - Designed a custom PCB for ESP32 with an Arduino Mega footprint for standardized micro-controller testing - Utilized FreeRTOS to interface with peripherals and validate hardware-software integration.	Jan. 2023 – May 2023

EXPERIENCE

Vertically Integrated Projects <i>Purdue University</i> - Developed Reinforcement Learning algorithms for a robotic UR5e hand to perform cello bowing based on motion capture data. - Simulated kinematics and bow manipulation dynamics within a Gazebo environment to optimize control policy.	Aug. 2024 – Dec. 2024 <i>West Lafayette, IN</i>
Machine Learning Intern <i>KAYA Global, INC.</i> - Designed a chatbot context system with a Node backend by integrating custom Llama2 models via LangChain. - Deployed cloud services using GCP Cloud Functions for transcription and context extraction with CI/CD.	May 2023 – Dec. 2023 <i>New York City, NY</i>

TECHNICAL SKILLS

Languages: C/C++, Python, SystemVerilog, Assembly (ARM/RISC-V), JavaScript
Robotics: ROS/ROS2, Gazebo, PyBullet, SLAM, MPC, RRT*, PID Control
Tools: CI/CD, Linux (Ubuntu), Git, Docker, OpenCV, PyTorch, Raspberry Pi, ESP32
Hardware: PCB Design (EAGLE), I2C, SPI, FPGA Synthesis